

Bacteria transformation

Overview

Introduce foreign plasmid DNA in competent cells in order to replicate plasmids in *E. coli*.

A. Materials

- a. SOC medium
 - i. LB medium
 - ii. 0.5 M sorbitol
 - iii. 0.38 M mannitol
- b. Competent cells
 - i. DH5 α competent cells
- c. Plasmid
- d. LB plate
 - i. With Ampicillin for level 1 construct
 - ii. With Chloramphenicol for level 0 and 2 constructs
- e. Ice bucket with ice
- f. Water bath
- g. Cell spreader
- h. Bunsen burner

B. Procedure

1. Set the water bath to 42°C and prepare an ice bath.
2. Get plates (depending on antibiotics selection) and pre-warm them at 37°C.
3. Thaw competent cells on ice until they are 50% thawed and add 10 μ l of plasmid to 100 μ l of competent cells.
4. Incubate on ice for 20 min
5. Heat shock 42°C for 45 seconds
6. Incubate on ice for 3 min
7. Add 1 mL SOC medium
8. Incubate for 1 hr at 37°C
9. Centrifuge at 5000xg for 5 min
10. Remove ~1 mL supernatant and resuspend
11. Spread on plate
12. Incubate at 37°C for 16 hours

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